

Updated Project Proposal: Weather and Bicycle Counts in Munich

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Dataset

We chose a dataset about daily weather and bike counts in Munich.

- **Source:** We found it on the OpenData München website.
- **License:** It is free to use (dl-de-by-2.0 license).
- **Size:** The CSV files are under 10 MB, which meets the project requirement.
- **Variables we use:**
 - **Dependent Variable:** gesamt (Total bikes per day).
 - **Time Variables:** datum (Date, which we use to extract Month and Year).
 - **Weather Variables:** max . temp (Highest temperature) and niederschlag (Precipitation/Rain).

The dataset is still the same as in our first proposal. We use data from 2021, 2022, 2023 and 2024. This allows us to compare bicycle counts across different months and years.

Research Questions

Our research questions are also still the same:

- 1 How do daily bicycle counts in Munich vary across seasons and years?
- 2 How are daily weather conditions, especially temperature and precipitation, related to bicycle counts?

For the updated proposal, we also want to think more about how we can show our results clearly in the final project.

Progress since the first proposal

Since the first project proposal, we have started to work more with the data.

We have already done these steps:

- imported the CSV files for 2021, 2022, 2023 and 2024
- combined the four years into one dataset

- selected the most important variables
- renamed the variables into clearer names
- changed the date variable into a date format
- changed the bike counts, temperature and precipitation variables into numeric format
- created month and year variables
- removed missing values in the variables we use
- made first plots for our research questions

The main data cleaning is mostly done. We are now moving more to visualisation, interpretation and presentation.

Initial EDA

```
#| label: load-data
#| message: false
#| warning: false

library(tidyverse)
library(lubridate)
library(here)

# 1. Read all files as character to avoid format problems
data_2021 <- read_csv(here("data", "bike_2021.csv"), col_types = cols(.default = "c"))
data_2022 <- read_csv(here("data", "bike_2022.csv"), col_types = cols(.default = "c"))
data_2023 <- read_csv(here("data", "bike_2023.csv"), col_types = cols(.default = "c"))
data_2024 <- read_csv(here("data", "bike_2024.csv"), col_types = cols(.default = "c"))

# 2. Combine the four years together
bike_data_raw <- bind_rows(data_2021, data_2022, data_2023, data_2024)

# 3. Clean data: select, rename, and convert variable types
bike_data <- bike_data_raw %>%
  select(datum, gesamt, max.temp, niederschlag) %>%
  rename(
    date = datum,
    total_bikes = gesamt,
    max_temp = max.temp,
    precipitation = niederschlag
  ) %>%
  mutate(
    date = ymd(date),
    total_bikes = as.numeric(total_bikes),
    max_temp = as.numeric(max_temp),
    precipitation = as.numeric(precipitation),
    month = month(date, label = TRUE),
    year = as.factor(year(date))
  ) %>%
  drop_na()
```

First plot: Bicycle counts by month

```
#| label: bikes-by-month
#| message: false
#| warning: false

ggplot(bike_data, aes(x = month, y = total_bikes, fill = month)) +
  geom_boxplot(show.legend = FALSE) +
```

```
theme_minimal() +
labs(
  title = "Bicycle Counts Across Months",
  x = "Month",
  y = "Total Bikes"
)
```

This plot helps us answer the first research question. It shows that bicycle counts are different across the year. There are more bicycles in warmer months and fewer bicycles in colder months.

Second plot: Temperature and bicycle counts

```
#| label: temperature-and-bikes
#| message: false
#| warning: false

ggplot(bike_data, aes(x = max_temp, y = total_bikes)) +
  geom_point(alpha = 0.3, color = "green") +
  geom_smooth(method = "lm", color = "orange") +
  theme_minimal() +
  labs(
    title = "Temperature and Bicycle Counts",
    x = "Max Temperature (C)",
    y = "Total Bikes"
  )
```

This plot helps us answer the second research question. It shows that higher temperature is related to higher bicycle counts. This means that more people seem to cycle when the weather is warmer.

What we found so far

Our first results are:

- Bicycle counts change a lot across the year.
- Bicycle counts are higher in warmer months.
- Bicycle counts are lower in colder months.
- Higher temperature seems to be related to more bicycle counts.

These findings are still preliminary. We will improve the plots and explanations for the final submission.

Updated Analysis Plan

For the final project, we want to keep the analysis simple and clear.

Our plan is:

- 1 **Data Cleaning:** We will check the cleaned dataset again and make sure the code works from beginning to end.
- 2 **Analysis for Q1: Time and Seasons:** We will use plots and summary statistics to show how bicycle counts change by month and year.

- 3 **Analysis for Q2: Weather:** We will focus on the relationship between bicycle counts and temperature, while using precipitation only as a supporting weather variable if it helps the interpretation.
- 4 **Simple Model:** If possible, we will use a simple linear regression model with `total_bikes` as the dependent variable. The main independent variable will be `max_temp`, and we may add year or precipitation as controls.
- 5 **Presentation:** We will focus on making the plots easy to understand. We will add clear titles, axis labels and short explanations.

Communication and visualisation plan

The next step is not only analysis, but also communication and presentation.

For the final report, we want to:

- introduce the topic and the dataset
- explain the research questions
- show the most important plots
- write short explanations for each plot
- avoid making the report too complicated
- make the final document easy to read
- render the final project as a webpage or HTML document

We want the reader to understand the main results without needing much statistical knowledge.

Remaining tasks

Before the final submission, we still need to:

- improve the plots
- decide which plots are most important
- write better explanations for the results
- check the regression model
- check that the code is reproducible
- render the final document
- check the GitHub repository
- write the group reflection
- prepare the individual contribution statements

Conclusion

Our project focuses on how daily bicycle counts in Munich change across seasons and how weather, especially temperature, is related to cycling activity. So far, the cleaned data and first plots suggest a clear seasonal pattern: bicycle counts are higher in warmer months and lower in colder months. The temperature plot also supports the idea that warmer days are associated with more bicycle trips.

For the final report, we will focus on explaining these two main results clearly. We will use the monthly bicycle-count plot and the temperature plot as our key visuals, improve their labels and interpretation, and connect them directly to the research questions. This will make the final project more focused and easier for readers to understand.